

Zomato Delivery Operations Analytics

Delivery Performance & Resource Optimization

Business Analyst Portfolio Project 2

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MYSQL

POWER BI

RCA

EXECUTIVE SUMMARY

Business Problem

Why was this project needed?

Inconsistent Delivery Times

Inconsistent delivery times affecting customer experience

Unclear SLA Drivers

Unclear drivers of SLA breaches

Uneven Rider Workload

Uneven rider workload distribution

Resource Allocation Imbalance

Resource allocation imbalance across cities

 **Business Need:** Identify the root causes of delivery delays and recommend operational improvements using historical order data.

Objectives



**Reduce Average
Delivery Time**



**Optimize Rider
Performance**



**Optimize Resource
Allocation Across Cities**



**Improve Customer
Experience (SLA-Based
Proxy)**



**Understand Operational
Bottlenecks**

Project Overview

Dataset: Third-party operational dataset (Kaggle)

Dataset Details

- 45,585 Rows
- 22 Columns
- 3 Tables

Tools & KPIs

Tool Used: MySQL | Power BI

KPI: 6 Key KPIs developed

Data Quality Findings

Impact:

Missing Values

Missing values in rider age, rating, and timestamps (4–13% of records)

Invalid Values

Invalid values: ratings up to 6.0 (outside 1–5 scale), rider ages as low as 15

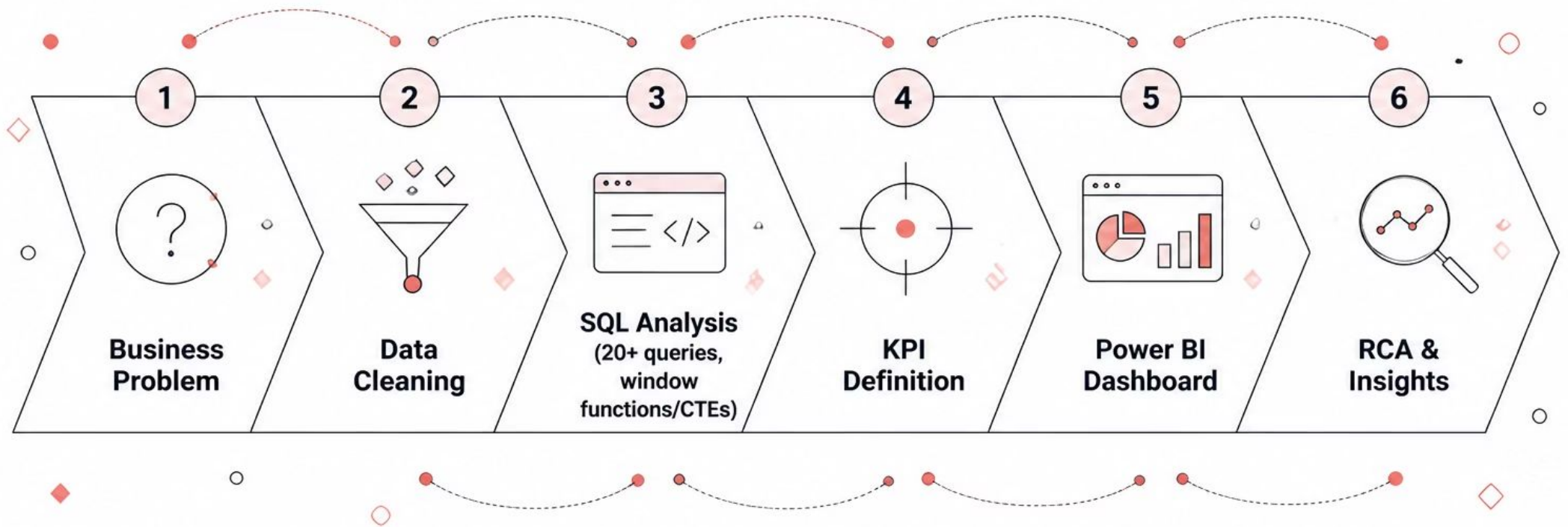
Unreliable Preparation-Time Fields

Preparation-time fields unreliable — negative values and outliers up to 1,435 minutes

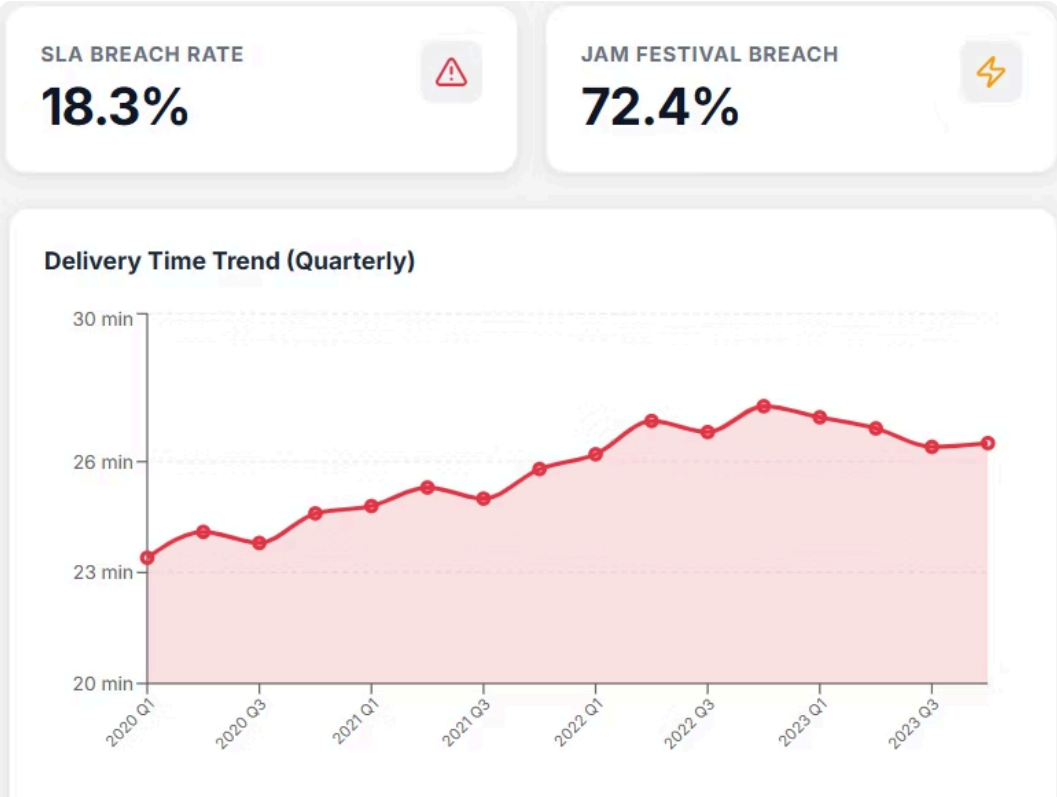
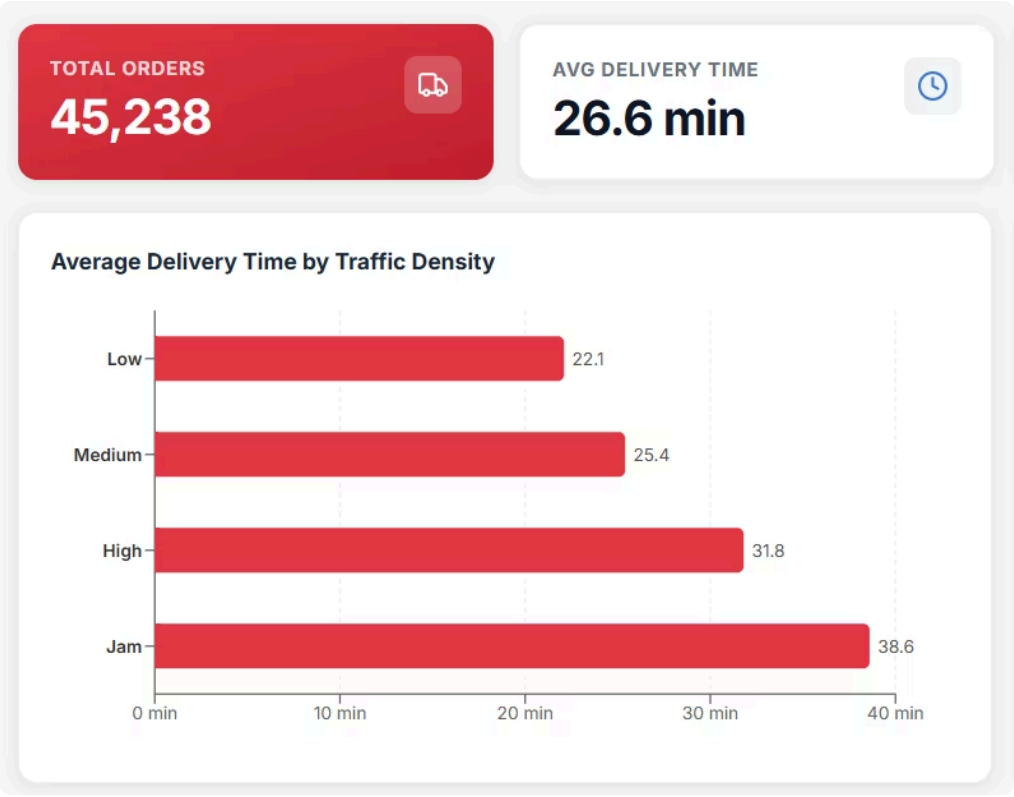
Scoped Analysis

No customer ID or order value in source data — scoped analysis to delivery ops, not revenue/retention

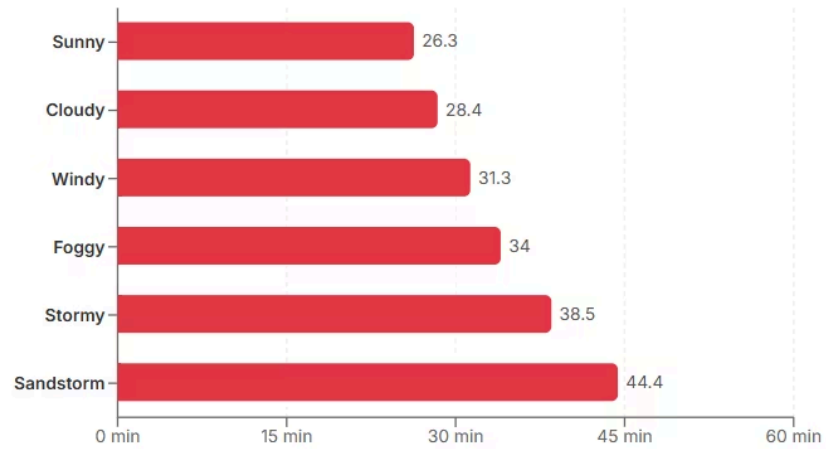
Analysis Process



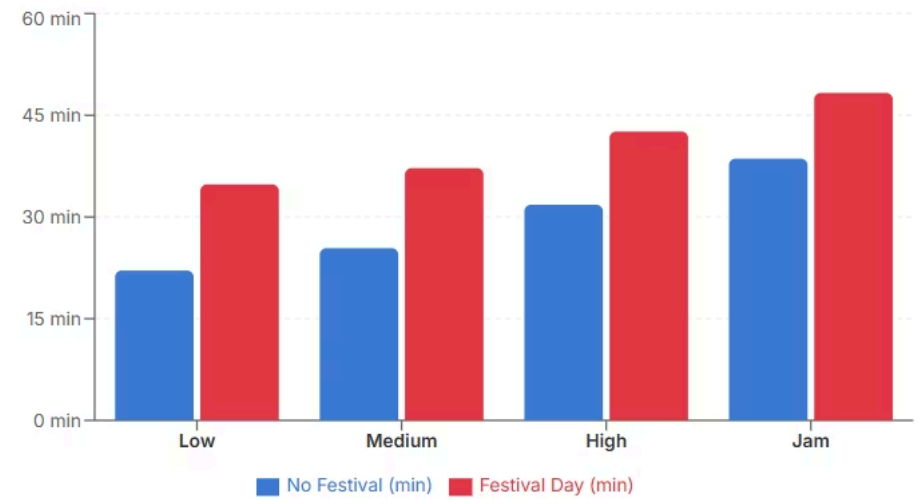
Dashboard



Average Delivery Time by Weather Condition



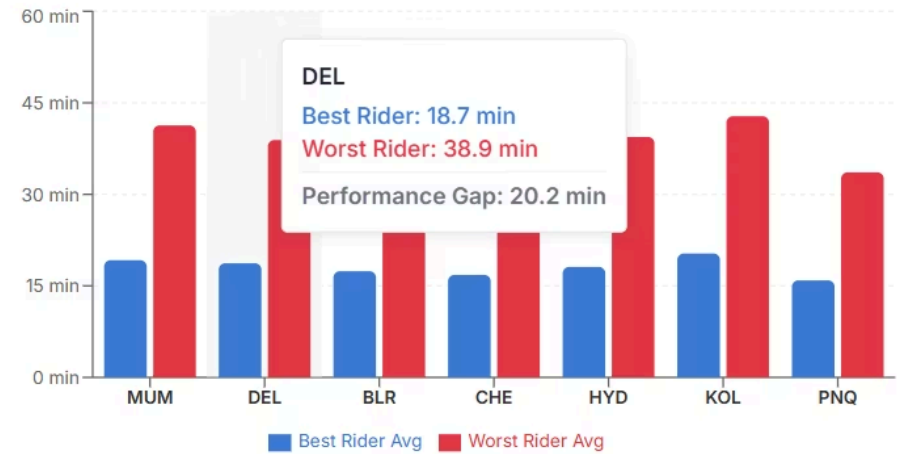
Average Delivery Time by Festival



Order Volume by City



Best Rider Avg vs Worst Rider Avg and Performance Gap by City



Average Delivery Time by Number of Simultaneous Deliveries

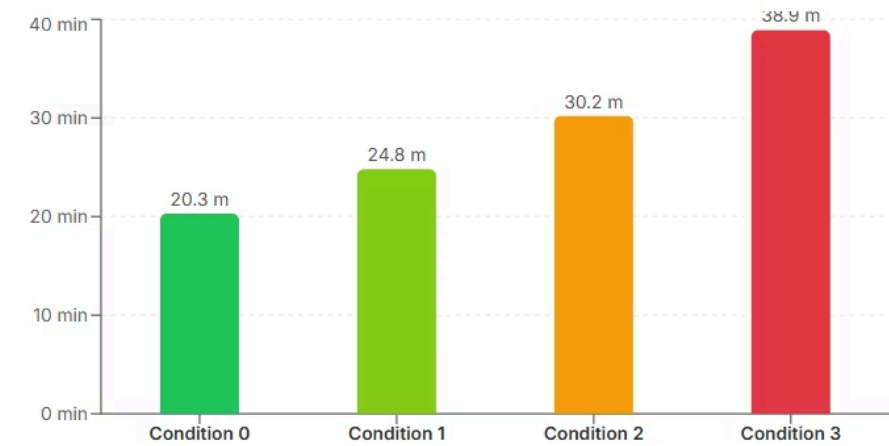


45.3 min

AVG TIME: 3 DELIVERIES

Filtered by current selection

Average Delivery Time by Vehicle Condition



Delivery Time (min): Multiple Orders x Vehicle Condition

Orders / Condition	Cond 0	Cond 1	Cond 2	Cond 3
1 Order	19.2	22.4	26.8	32.1
2 Orders	24.1	27.8	31.4	40.2
3 Orders	29.6	34.2	38.7	46.8

Green <25 min · Amber 25~35 min · Red >35 min

Key Insights

91.14%

On-Time Delivery Rate

47.82

Min Avg Delivery Time

at max batching (3 deliveries) vs. 22.88 min at zero

92.06%

SLA Breach Rate

on Jam+Festival days

74.8%

Metro City Order Volume

of order volume concentrated in metro cities

Root Cause Highlights

Rider workload (batching)

Largest single controllable factor order age, rating, and timestamps (4–13% of records)

Festival + traffic compounding

Not additive, multiplicative

Fleet + workload compounding

Worst recorded case at 43.34 mins

Corrected finding

City-level rider gap is a sample-size artifact, not a skill gapscored analysis to delivery ops, not revenue/retention

Recommendations

1

Cap batching at one delivery per rider

2

Build a festival-specific staffing plan

3

Restrict multi-delivery assignments for poor-condition vehicles

4

Prioritize rider capacity in metro hubs

Expected Business Impact

- Reduced SLA breaches

- More predictable delivery times

- Better rider workload balance

- Smarter festival-day resourcing

- Improved fleet utilization

- Stronger data reliability for future analysis

Business Analyst Skills

Business Analysis:

- Requirement Gathering,
- Root Cause Analysis,
- KPI Definition,
- Recommendations

Data Analysis:

- MySQL (window functions, CTEs)
- Power BI
- DAX
- Data Cleaning

Deliverables:

RCA & KPI Framework

Executive Summary

Power BI Dashboar

Business Presentation